

## Dr. HONGPEI YANG

School of Atmospheric Sciences, Sun Yat-sen University

Personal Website: <https://hongpei-yang.github.io/>

E-mail: [yanghp731@gmail.com](mailto:yanghp731@gmail.com), [yanghp9@mail2.sysu.edu.cn](mailto:yanghp9@mail2.sysu.edu.cn)



[ORCID](#)



[ResearchGate](#)



[Google Scholar](#)



[LinkedIn](#)

### PROFESSIONAL EXPERIENCE

01/08/2026–  
present      **Postdoctoral Research Fellow**, Sun Yat-sen University, China  
Host: Prof. Guixing Chen & Prof. Yu Du

### EDUCATION

01/09/2021–  
17/06/2026      **Ph.D. in Atmospheric Sciences**, Sun Yat-sen University, China  
Supervisor: Prof. Yu Du  
Thesis title: *Generation of Convective Gravity Waves and Their Impacts on Convection*

04/05/2023–  
27/07/2023      **Visiting Ph.D. student**, Mesoscale and Microscale Meteorology Laboratory, NSF  
National Center of Atmospheric Research, USA  
Host: Dr. Juanzhen (Jenny) Sun

01/09/2017–  
20/06/2021      **B.Sc. in Atmospheric Sciences**, Nanjing University of Information Science &  
Technology, China  
Supervisor: Dr. Jiaping Xue and Prof. Jingjia Luo  
Thesis title: *Evaluation of Simulated Ningaloo Niño/Niña in CMIP6 Models*

### HONORS and AWARDS

#### *Academic Honours*

- **Postdoctoral Research Fellowship**, Sun Yat-sen University, China (2026)
- **Outstanding Doctoral Dissertation Award**, Sun Yat-sen University, China (2026)
- **Outstanding Doctoral Graduate**, Sun Yat-sen University, China (2026)
- **China National Scholarship for Doctoral Students**, Ministry of Education, China (2025)
- **President's Scholarship**, Sun Yat-sen University, China (2025)
- **Outstanding Undergraduate Graduate**, Nanjing University of Information Science & Technology, China (2021)

#### *Conference Presentation Awards*

- **Gold Award**, "Three-Minute Thesis" Competition, Sun Yat-sen University, China (2026)
- **Outstanding Oral Presentation Award**, 2nd Future Atmospheric Science Forum, Suzhou, China (2026)
- **First Prize Oral Presentation Award**, 1st Navier–Stokes Doctoral Student Academic Forum, Fudan University, Shanghai, China (2025)
- **Best Oral Presentation Award**, 7th National Forum on Mesoscale Meteorology, Datong, China (2025)
- **Best Student Poster Award**, 16th International Conference on Mesoscale Convective Systems (ICMCS-XVI), Gyeongju, South Korea (2024)
- **Excellent Student Presentation Award**, 6th National Conference on Mesoscale Meteorology, Zhuhai, China (2024)
- **Outstanding Student Presentation Award**, 15th International Conference on Mesoscale Convective Systems (ICMCS-XV), Fort Collins, USA (2023)

**RESEARCH GRANTS**

- [1] **National Natural Science Foundation of China** – Basic Research Program for PhD Students  
**Principal Investigator**, Grant No. 424B200063, 01/2025 – 12/2027, CNY 300,000  
 Title: *A Comparative Study of the Mechanisms by Which Two Types of Convection-Triggered Gravity Waves Influence Severe Convection*
- [2] **Jiangsu Provincial College Students' Innovation and Entrepreneurship Training Program**  
**Principal Investigator**, Grant No. 202010300144, 04/2020 – 03/2021, CNY 3,000  
 Title: *Formation Mechanisms of Heavy Rainfall during Anomalous Meiyu Seasons*

**REFEREED PUBLICATIONS**

- [1] **Yang H.**, Y. Du, and J. Wei, 2023: Generation of multiple gravity wave couplets from convection. *J. Atmos. Sci.*, 80, 2323–2343.
- [2] **Yang H.**, and Y. Du, 2024: Difference between upshear- and downshear-propagating waves associated with the development of squall lines. *Mon. Wea. Rev.*, 152, 1399–1420.
- [3] **Yang H.**, Y. Du, Z. Chen, and J. Fang, 2024: Could developing frontal rainfall influence warm-sector rainfall? *Geophys. Res. Lett.*, 51, e2024GL110430.
- [4] **Yang H.**, Y. Du and J. Sun, 2025a: The merger of supercell and squall line in the Great Plains. 1: Initiation of the supercell. *J. Geophys. Res. Atmos.*, 130, e2024JD042393.
- [5] **Yang H.**, Y. Du and J. Sun, 2025b: The merger of supercell and squall line in the Great Plains. 2: Formation of bow echo. *J. Geophys. Res. Atmos.*, 130, e2024JD042394.
- [6] **Yang H.**, and Y. Du, 2026: Distinct convection initiation near and far ahead of an idealized squall line. *J. Atmos. Sci.*, 83, 151–168.
- [7] **Yang H.**, Y. Du, Z. Chen, and X. Gao, 2026: Convection initiation over mountain slopes in North China: Roles of upslope winds and orographic waves. *Adv. Atmos. Sci.*, 43, 889–906.
- [8] Xue J., **H. Yang\***, J. Luo, C. Yuan, B. Wang, and T. Yamagata, 2022: Ningaloo Niño/Niña in CMIP6 models: Characteristics, mechanisms, and climate impacts. *Geophys. Res. Lett.*, 49, e2022GL099781.
- [9] Zhou X., Y. Du, J. Wei, Z. Chen, and **H. Yang**, 2024: Statistical characteristics of wavelike banded convection associated with ducted gravity waves over southern China. *Geophys. Res. Lett.*, 51, e2024GL112027.
- [10] Du Y., R. Rotunno, Z. Chen, and **H. Yang**, 2024: A linear theory for periodic convectively forced gravity waves near a coastline. *J. Atmos. Sci.*, 81, 1271–1288.
- [11] Z. Chen, Y. Du, C. Vincent, E. Short and **H. Yang**, 2024: Influence of coastal topography on offshore diurnal rainfall propagation dynamics: A linear gravity wave model approach. *J. Atmos. Sci.*, 82, 1911–1927.
- [12] Fu D., Y. Du, and **H. Yang**, 2026: The interaction between low-level jets and cold pools and their impacts on convection. *J. Atmos. Sci.*, 83, 861–879.
- [13] Chen Z., Y. Du, C. Vincent, E. Short, **H. Yang**, and G. Chen, 2026: What causes faster or slower offshore propagation of diurnal rainfall in New Guinea? *J. Atmos. Sci.*, 83, 1207–1224.

*In revision*

- [1] Chen Z., Y. Du, C. Vincent, E. Short, **H. Yang**, and G. Chen, 2026: Distinguishing gravity waves forced by land-sea thermal contrast and convection in offshore rainfall propagation. *J. Atmos. Sci.* pending revisions.
- [2] Shao J., H. Yang, and Y. Du, 2026: The combined effects of convective and orographic gravity waves on convection initiation. *J. Atmos. Sci.*, pending revisions.

**INVITED SEMINAR/TALKS**

- [1] **High-resolution chat**, European Centre for Medium-Range Weather Forecasts (ECMWF), Online (2026)  
*How do convectively generated gravity waves affect convection initiation?*

- [2] **Invited talk**, 2nd Future Atmospheric Science Forum, Suzhou, China (2026)  
*How do convective gravity waves influence convection initiation and its spatiotemporal characteristics?*
- [3] **Seminar**, Mesoscale & Microscale Meteorology Laboratory, National Center for Atmospheric Research, Boulder, USA (2023).  
*Supercells ahead of squall lines and their merger in the Great Plains.*

### **INTERNATIONAL CONFERENCE PRESENTATIONS**

- [1] **Oral presentation**, 21st Conference on Mesoscale Processes (AMS), Boise, USA (2025).  
*Cold pool and gravity waves drive convection initiation ahead of squall lines.*
- [2] **Oral presentation**, EGU General Assembly 2025, Vienna, Austria (2025).  
*Cold pool and gravity waves drive convection initiation ahead of squall lines.*
- [3] **Oral presentation**, 20th Annual Meeting of the Asia Oceania Geosciences Society (AOGS), Singapore (2023).  
*The role of gravity waves in the asymmetric development of squall lines.*
- [4] **Oral presentation**, 20th Conference on Mesoscale Processes (AMS), Madison, USA (2023).  
*Gravity waves associated with frontal rainfall as a preconditioning mechanism for warm-sector heavy rainfall in South China.*
- [5] **Oral presentation**, 3rd Symposium on Mesoscale Processes, 103rd AMS Annual Meeting, Denver, USA (2023).  
*The role of gravity waves in the asymmetric development of mesoscale convective systems.*
- [6] **Poster presentation**, 16th International Conference on Mesoscale Convective Systems (ICMCS-XVI), Gyeongju, South Korea (2024).  
*Bow echo resulting from squall line–supercell merger in the Great Plains of the United States.*
- [7] **Poster presentation**, 15th International Conference on Mesoscale Convective Systems (ICMCS-XV), Fort Collins, USA (2023).  
*Gravity waves associated with frontal rainfall as a preconditioning mechanism for warm-sector heavy rainfall in South China.*
- [8] **Poster presentation**, AGU Fall Meeting 2022, Chicago, USA (2022).  
*The role of gravity waves in the asymmetric development of mesoscale convective systems.*

### **TEACHING EXPERIENCE**

**Teaching Assistant**, *Dynamical Meteorology* (Undergraduate Course), Sun Yat-sen University, 09/2022 – 01/2023, assisted with undergraduate teaching, including problem-solving classes, tutorials, and student discussions.

### **ACADEMICS SERVICES**

- [1] **Manager and Editor, WeChat Official Account “Weather Watch”** (08/2024 – present)  
Key responsibilities: Inviting researchers in atmospheric science to share their latest research and edit articles to communicate scientific findings in accessible language.  
Account statistics (as of 08/2026): 182 posts published; 11,683 subscribers.
- [2] **Reviewer** for International Journals, including:  
*Nature Communications; Journal of Fluid Mechanics; Journal of the Atmospheric Sciences; Monthly Weather Review; Atmospheric Chemistry and Physics; Quarterly Journal of the Royal Meteorological Society; Geophysical Research Letters; Journal of Geophysical Research: Atmospheres; Advances in Atmospheric Sciences; Atmospheric Science Letters; npj Climate and Atmospheric Science, Climate Dynamics; Meteorological Applications; Atmospheric Research, Asia-Pacific Journal of Atmospheric Sciences.*